4 An electric train of mass 1.5×10^4 kg travels up a slope at a constant speed as shown in Figure 4.1.

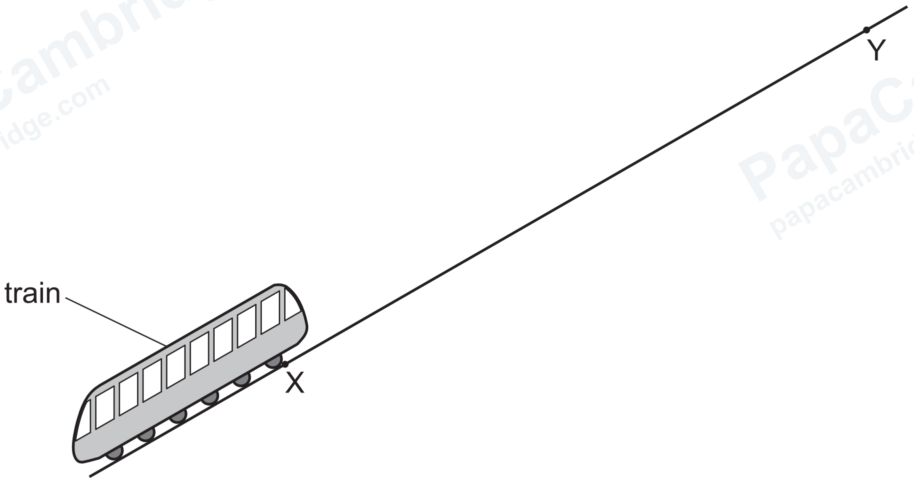

Figure 4.1

At point X, the train has gravitational potential energy 7.4 MJ.
At point Y, the train has gravitational potential energy 13.0 MJ.
The train takes a time of 52 s to move from X to Y.
The kinetic energy of the train is 1.3×10^5 J.

(a) Calculate the speed of the train.

speed = ... $m\,s^{-1}$ [3]

(b) The train is powered by an electric motor. The useful output power of the motor is 1.1×10^5 W.

(i) Define power.

.... [1]

(ii) Determine the resistive force F acting on the train as it moves from point X to point Y.

F = ... N [3]

(iii) The current in the motor is 95 A. The potential difference (p.d.) across the motor is 1400 V.

Calculate the percentage efficiency of the motor.

percentage efficiency = ... % [3]

[Total: 10]